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Foreword

This ETSI Technical Report (ETR) has been produced by the Special Mobile Group (SMG) Technical Committee (TC) of the European Telecommunications Standards Institute (ETSI).

This ETR describes how interworking can be obtained between phase 2 mobile stations and phase 1 infrastructure of the European digital cellular telecommunications system (phase 2).

This ETR is an informative document resulting from SMG studies which are related to the European digital cellular telecommunications system (phase 2). This ETR is used to publish material which is of an informative nature, relating to the use or the application of ETSs and is not suitable for formal adoption as an ETS.

This ETR correspond to GSM technical specification, GSM 09.90 version 4.1.0.

Reference is made within this ETR to GSM Technical Specifications (GSM-TS) (NOTE).

NOTE: TC-SMG has produced documents which give the technical specifications for the implementation of the European digital cellular telecommunications system. Historically, these documents have been identified as GSM Technical Specifications (GSM-TS). These TSs may have subsequently become I-ETSs (Phase 1), or ETSs (Phase 2), whilst others may become ETSI Technical Reports (ETRs). GSM-TSs are, for editorial reasons, still referred to in current GSM ETSs.

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1. Scope.

The Scope of this Technical Report is to clarify how interworking can be obtained between phase 2 mobile stations and phase 1 infrastructure. The objective is to obtain this without changing the consolidated set of phase 1 specifications. This Technical Report specifies the necessary amendments to the phase 1 infrastructure so that an acceptable service is offered to mobile stations of phase 2, guaranteeing that a phase 2 mobile station obtains all phase 1 services.

This report outlines the necessary changes as well as the necessary controls and clarifications with regard to phase 1 implementation, which should be performed to the different interfaces in phase 1 infrastructure before phase 2 mobiles are offered service.

2. References.

This ETR incorporates by dated and undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this ETR only when incorporated in it by amendment or revision. For undated references, the latest edition of the publication referred to applies.

[1] GSM 01.04: "European digital cellular telecommunication system (Phase 2); Abbreviations and acronyms".

[2] GSM 04.06: "European digital cellular telecommunication system (Phase 2); Mobile Station - Base Station System (MS - BSS) interface Data Link (DL) layer specification".

[3] GSM 04.08: "European digital cellular telecommunication system (Phase 2); Mobile radio interface layer 3 specification".

[4] GSM 04.11: "European digital cellular telecommunication system (Phase 2); Point-to-Point (PP) Short Message Service (SMS) support on mobile radio interface".

[5] GSM 04.06 Phase 1 (I-ETS 300 021):"European digital cellular telecommunication system (Phase 1); Mobile Station - Base Station System (MS - BSS) interface Data Link (DL) layer specification".

[6] GSM 04.08 Phase 1 (I-ETS 300 022-1): "European digital cellular telecommunication system (Phase 1); Mobile radio interface layer 3 specification Part 1: Generic".

[7] GSM 04.08-DCS Phase 1 (I-ETS 300 022-2): "European digital cellular telecommunication system (Phase 1); Mobile radio interface layer 3 specification Part 2: DCS extension".

[8] GSM 04.08-EXT Phase 1 (I-ETS 300 022-3): "European digital cellular telecommunication system (Phase 1); Mobile radio interface layer 3 specification Part 3: Signalling support of the second ciphering algorithm".

[9] GSM 04.11 Phase 1 (I-ETS 300 023): Point-to-point short message service support on mobile radio interface".

2.1. Definitions and abbreviations.

Abbreviations used in this document are listed in GSM 01.04.

3. General.

To obtain compatibility between phase 2 mobile stations and phase 1 infrastructure, it has been attempted to design phase 2 as an extension of the phase 1 protocols. Basically meaning that the phase 1 protocols are contained as a subset of the phase 2 protocols and that for most phase 1 functionality, phase 1 and phase 2 signalling are identical.

This approach in general has been followed. However the requirement for introduction of the new phase 2 features has in some cases required deviation from this general rule. Furthermore, due to freedom in the phase 1 technical specifications there might be different phase 1 infrastructure implementations. Therefore it has not in all cases been possible to ensure interworking with all potential implementations.

The remainder of this Technical Report describes how to overcome the possible impacts of the above mentioned factors.

In section 4 the unavoidable service restrictions a phase 2 mobile station will encounter in phase 1 infrastructure are explained. Sections 5, 6, 7 and 8 give general and specific behaviour of phase 1 implementations which has been assumed and as a result must be ensured. The unavoidable changes and additions to phase 1 infrastructure needed to obtain proper working of a phase 2 mobile station in phase 1 infrastructure is also given.

Within the scope of this ETR the following definitions are used :

- a phase 1 mobile station is a mobile station which is compliant with the protocols defined by the phase 1 specifications;

- a phase 2 mobile station is a mobile station which is compliant with the protocols defined by the phase 2 specifications;

- a phase 1 infrastructure is an infrastructure which is compliant with the protocols defined by the phase 1 specifications;

- an upgraded phase 1 infrastructure is a phase 1 infrastructure which has been upgraded according to this ETR.

4. Service restrictions.

When the specifying part of this Technical Report is implemented and an upgraded phase 1 infrastructure has been created, a phase 2 mobile station will have no service restrictions for the features and services defined for phase 1. However, no means have been taken to provide a phase 2 mobile station in an upgraded phase 1 infrastructure, with the features and services added in the phase 2 standard compared to the phase 1 standard.

5. Air interface.

This section deals with enhancements of the protocols at the air interface between phase 1 and phase 2 of GSM. The protocols concerned are defined in TS GSM 04.06, TS GSM 04.08 and TS GSM 04.11; the only paragraph for which TS GSM 04.06 is concerned is in section 5.1.1 of this ETR and tackles the treatment of spare bits.

For each change from phase 1 to phase 2, explanations are provided; this informative part is written in roman characters. Then a specifying part is provided which is written in italic characters. These specifications are those which are absolutely necessary for the service provision to a phase 2 mobile station. In some cases, additional specifications of desirable upgrades to accommodate enhanced phase 2 mobile stations are provided.

5.1 General behaviour of upgraded phase 1 infrastructures.

5.1.1 Spare bits.

The following rules about spare bits apply to both protocols defined in TS GSM 04.06 and 04.08 for the phase 1 and phase 2 specifications.

Where the description of information elements or messages in the phase 1 specifications contains spare bits, these bits are indicated as being set to "0". Moreover, enhancements of the protocols have been done on the assumption that spare bits are ignored.

To ensure compatible evolution of the protocols, an upgraded phase 1 infrastructure shall set spare bits to "0".

An upgraded phase 1 infrastructure is not allowed to reject messages simply because a bit which is defined as spare in the phase 1 specifications and used by the phase 2 specifications, is set to "1".

In addition, it would be advisable that an upgraded phase 1 infrastructure does not reject messages simply because a bit which is defined as spare in the phase 1 specifications is set to "1".

Section 5.2.6 summarizes the phase 2 changes to TS GSM 04.08 which have made use of the phase 1 spare bits.

5.1.2 Values "reserved for future use".

In the phase 1 definition of information elements, some data fields may contain values that are defined as "reserved" or "reserved for future use".

Except for those cases which are indicated in sections 5.2.1.1, 5.2.1.2, 5.2.4.2, and 5.2.5.1 of this ETR, if a value defined as "reserved" in the phase 1 specifications is detected by an upgraded phase 1 infrastructure in an information element, this shall be considered as an error and the action yielded by this error shall be taken according to the error handling already implemented.

5.2 Specific behaviour of upgraded phase 1 infrastructures.

This section deals with the changes which have been introduced to TS GSM 04.08 and TS GSM 04.11 for phase 2 and which have impact on the interworking between phase 1 infrastructure and phase 2 mobile stations.

5.2.1 Radio resource procedures.

5.2.1.1 RR-connection establishment.

New values for the field "establishment cause" of the CHANNEL REQUEST message have been introduced to improve the RR-connection establishment procedure. However there will be no problem because a phase 1 infrastructure does not indicate support of new establishment causes nor add any paging indication in the PAGING REQUEST messages; hence a phase 2 mobile station uses only the phase 1 values when sending CHANNEL REQUEST messages. For mobile originating SS and SMS the phase 2 MS will use the establishment cause used for location updating in phase 1.

An upgraded phase 1 infrastructure shall accept that an MS requires SS or SMS after it had used "000" as establishment cause.

To foresee the future enhancements of the CHANNEL REQUEST message, an upgraded phase 1 infrastructure should not reject a CHANNEL REQUEST message simply because the "establishment cause" field is coded with a reserved value; moreover, when receiving such a reserved establishment cause value, an upgraded phase 1 infrastructure should allocate an SDCCH if it is not able to determine the mobile station capabilities.

If the above advice is taken into account, this case is one of the exceptions to section 5.1.2 of this ETR.

5.2.1.2 Assignment and handover procedures.

After reception of ASSIGNMENT COMMAND or HANDOVER COMMAND messages the mobile station may send ASSIGNMENT FAILURE or HANDOVER FAILURE messages containing new error causes : "channel mode unacceptable" or "frequency not implemented".
This ETR does not cover the recovery mechanisms in cases where the infrastructure has made errors. However,

An upgraded phase 1 infrastructure is not allowed to consider as erroneous or invalid ASSIGNMENT FAILURE or HANDOVER FAILURE messages containing one of the following error causes : "channel mode unacceptable" or "frequency not implemented".

It is up to the upgraded phase 1 infrastructure whether or not to treat these new error causes as the error causes already defined in the phase 1 specifications.

This case is one of the exceptions to section 5.1.2 of this ETR.

5.2.1.3 Transmission mode change.

On reception of a CHANNEL MODE MODIFY message, if a phase 2 mobile station finds that it can not change to the indicated new mode, it shall retain the old mode and return a CHANNEL MODE MODIFY ACKNOWLEDGE message including a "channel mode" information element describing the old mode.

An upgraded phase 1 infrastructure (which expects a description of the new mode) shall check whether the mode included in the CHANNEL MODE MODIFY ACKNOWLEDGE message is the mode that was ordered in the CHANNEL MODE MODIFY message, and if not it shall conclude to a failure of the transmission mode change procedure.

5.2.2 Mobility management procedures.

5.2.2.1 MM-connection establishment.

During the first mobile originated MM-connection establishment, a phase 2 mobile station can force a release before the completion of the establishment phase by sending a CM-SERVICE ABORT message. This message is a new one in phase 2, but in any case if the RR-connection is not released a phase 2 mobile station will abort the RR-connection at expiry of timer T3240.

An upgraded phase 1 infrastructure is not allowed to consider as erroneous or invalid a mobile station simply because it has sent a CM-SERVICE ABORT message. Hence, sending by a mobile station of a CM-SERVICE ABORT message will not cause the blacklisting of this mobile station.

5.2.2.2 Location updating procedure.

When a phase 2 mobile station is in service state LIMITED SERVICE of state MM-IDLE it is not obliged to respond to paging with IMSI. Under the same conditions a phase 1 mobile station shall respond to paging with IMSI.

An upgraded phase 1 infrastructure, after having rejected a location updating request with cause "Location area not allowed" or "PLMN not allowed" shall take into account the fact that phase 2 mobile stations need not respond when paged with IMSI.

The value "FFFE" is used in phase 2 for the mobile station to indicate a deleted LAC, for example after reception of a LOCATION UPDATING REJECT message with cause "location area not allowed".

An upgraded phase 1 infrastructure is not allowed to allocate value FFFE to describe one of its location areas, and shall consider value FFFE (for location area code) as meaning a deleted location area code.

5.2.2.3 Re-establishment procedure.

A phase 2 mobile station may indicate its IMEI as its mobile identity in a CM REESTABLISHMENT REQUEST message. It was not clear whether this is allowed on re-establishment for phase 1.

An upgraded phase 1 infrastructure is not allowed to consider as erroneous or invalid a mobile station simply because it has sent a CM REESTABLISHMENT REQUEST message including an IMEI as mobile identity.

5.2.3 Call control procedures.

5.2.3.1 Compatibility checking at call establishment.

New requirements have been introduced about compatibility checking when two "bearer capability" information elements are sent in the SETUP message.
The following procedure shall be taken into account by an upgraded phase 1 infrastructure : At the receipt of a SETUP message including two "bearer capability" information elements, a phase 2 mobile station may as a result of compatibility checking accept one bearer capability and reject the other one. In this case the phase 2 mobile station answers to the SETUP message with a CALL CONFIRMED message including the bearer capability which is accepted.

In those cases where one of the bearer capabilities provided to a phase 2 mobile station is not supported by this mobile station, an upgraded phase 1 infrastructure shall either release the call or establish the call according to the bearer capability supported by the mobile station.

5.2.3.2 Mobile originated call.

A phase 2 mobile station may include a new information element "CLIR suppression" (coded as a Type 2 information element) in the SETUP message to override the CLIR (Calling Line Identification Restriction) service.

If a "CLIR suppression" information element is included in the SETUP message, an upgraded phase 1 infrastructure shall process the SETUP message in a normal way.

In addition, an upgraded phase 1 infrastructure should either ignore or take into account the "CLIR suppression" information element.

5.2.3.3 Mobile terminating call.

In phase 1, the mobile station may send one or several "bearer capability" information elements in the CALL CONFIRMED message in the following two cases :

- the mobile station wishes another bearer capability than that given in the incoming SETUP message.
- the "bearer capability" information element is missing or not fully specified in the SETUP message.

In phase 2, the possibility is added for the mobile station to define any radio channel requirements (for example to indicate that the mobile station has half rate capability). Thus the "radio channel requirement" field of the "bearer capability" information element in the CALL CONFIRMED message may be different from that sent by the network in the SETUP message.
What follows concerns the cases where the "bearer capability" information element is acceptable by the network except for the radio channel requirement.

In case of mismatch of the radio channel requirement, if the received radio channel requirement indicates "00 (half-rate channel)" an upgraded phase 1 infrastructure is allowed to release the call; if there is another radio channel requirement (01, 10 or 11), an upgraded phase 1 infrastructure shall continue the call establishment.

In addition, an upgraded phase 1 infrastructure need not take into account the received radio channel requirement.

A phase 2 mobile station may include a new information element "connected subaddress" (coded as a Type 4 information element) in the CONNECT message.

If a "connected subaddress" information element is included in a CONNECT message, an upgraded phase 1 infrastructure shall process normally the CONNECT message.

In addition, an upgraded phase 1 infrastructure should either ignore or take into account the "connected subaddress" information element.

To avoid speech clipping a phase 2 mobile station will attach the user connection for speech at latest when sending the CONNECT message, except if there is no compatible radio resource available at this time.

An upgraded phase 1 infrastructure should ensure that the downlink speech path does not contain disturbing sound during the interval (if any) beginning at channel assignment and ending at through connection in the network. Typically, comfort noise or silence may be sent in this interval.

5.2.4 SMS procedures

5.2.4.1 Error Cause Mapping

Additional error causes have been specified for phase 2.

Where an error cause is not understood by the infrastructure it shall be treated as "unspecified error".

5.2.5 Messages.

5.2.5.1 General message coding.

In the phase 1 specifications it is stated that optional information elements of "types 1 and 2 may appear at any point among the optional information elements in a message" (cf par. 10.5 of TS GSM 04.08). In the phase 2 specifications, a general statement indicates that the order of appearance for information elements in a message is given by the table describing that message. Therefore a phase 2 mobile station expects these elements in that precise order.

An upgraded phase 1 infrastructure shall include optional type 1 and 2 information elements at their correct place in a message (this correct place is defined in the phase 2 table which describes the message).
This statement covers the following cases : optional "synchronisation indication" information element in HANDOVER COMMAND message; and optional "repeat indicator" information element in CALL CONFIRMED, CALL PROCEEDING and SETUP messages.

5.2.5.2 CLASSMARK CHANGE message.

A new information element "mobile station classmark 3" (coded as a Type 4 information element) may be included by a phase 2 mobile station in the CLASSMARK CHANGE message.

An upgraded phase 1 infrastructure is not allowed to consider as erroneous or invalid a mobile station simply because it has sent a CLASSMARK CHANGE message including a "mobile station classmark 3" information element.

5.2.5.3 RR-STATUS message.

The rewriting of error handling of the mobile station for phase 2 has introduced a new error cause value "invalid mandatory information" sent in the RR-STATUS message at detection of errors on mandatory information elements.
This ETR does not cover the recovery mechanisms in cases where the infrastructure has made errors. However,

An upgraded infrastructure is not allowed to consider as erroneous or invalid a mobile station simply because it has sent a RR-STATUS message with cause value "invalid mandatory information".

It is up to the upgraded phase 1 infrastructure to decide how to treat this new error cause.

This case is one of the exceptions to section 5.1.2 of this ETR.

5.2.5.4 STATUS message.

In TS GSM 04.08 section 5.5.3.2 it is made mandatory to release a call in some cases where a STATUS message is received. This can lead to very bad consequences as a call can be released when it could have been maintained. As a matter of fact the definition of "non implemented" being somehow ambiguous in phase 1, a phase 1 infrastructure having not implemented an information element or message can consider this signalling element a "non implemented" although it is defined in a phase 1 recommendation.

This ETR does not cover the recovery mechanisms in cases where the infrastructure has made errors, so it does not cover the cases when the mobile station sends a STATUS message after it has detected an error caused by the infrastructure.

An upgraded phase 1 infrastructure shall not send STATUS messages, except for the cases where the STATUS message is sent in response to a status enquiry procedure.

On reception of a STATUS message containing one of the following causes #96, #97, #99 or #100, a phase 1 mobile station shall clear the call but a phase 2 mobile station may decide not to clear the call. So, after having sent a STATUS message with one of the four causes above, the infrastructure can not rely on the mobile station clearing the call.

In those cases where, after having sent a STATUS message with one of the following causes #96, #97, #99 or #100, an upgraded phase 1 infrastructure aims to release the call, it should perform it explicitly after the sending of the STATUS message.

5.2.5.5 Messages relative to the SMS Point-to-Point

5.2.5.5.1 General

A phase 2 mobile may submit a short message containing in the Type-of-address a value for Type-of-Number or Numbering-Plan-Identification which is unknown to the SC.

Such short messages shall be rejected by a phase 1 SC with the error "Invalid SME address" conveyed by MAP, and forwarded by the VMSC to the MS as "SME transfer rejected".

5.2.5.5.2 CP-ERROR message.

In the phase 1 the CP-ERROR message is only used in the network to mobile station direction. In phase 2 the CP-ERROR message is used in both directions. Therefore a phase 2 mobile station may send a CP-ERROR message that is unexpected in phase 1 infrastructure.

An upgraded phase 1 infrastructure is not allowed to consider as erroneous or invalid a mobile station simply because it has sent a CP-ERROR message. This message shall simply be discarded. including an "auxiliary states" information element.

5.2.5.5.3 RP-ERROR message.

In the phase 2 the RP-ERROR message (GSM 04.11) can contain an additional (optional) information element "RP-User Data" which is used to convey failure causes between SC and MS. A phase 2 mobile station may send a RP-ERROR message, containing this element.

Upon receipt of a RP-ERROR message containing the parameter RP-User-Data, the upgraded phase 1 VMSC shall discard the information element RP-User-Data and then pass an error back to the SC as in phase 1, based on the value of RP-Cause from mobile.

5.2.5.5.4 SMS-SUBMIT

Within the SMS-SUBMIT, values for a number parameters have been extended in phase 2.

On receipt of an unknown value of any parameter, or use of bits in the first octet unspecified in phase 1, the upgraded phase 1 SC should reject the SMS-SUBMIT with the error "Invalid SME address" conveyed by MAP, and forwarded by the VMSC to the MS as "SME transfer rejected".
However, if only the Data Coding Scheme is set to a value unspecified in phase 1, and the SC is merely forwarding the message to another entity (i.e not performing any translation), the SC shall accept the message.

5.2.5.6 Supplementary Services messages.

A phase 2 mobile station may include a new information element "SS version indicator" (coded as a Type 4 information element) in some messages which contain a "Facility" information element, to aid the decoding of this "Facility" information element.

An upgraded phase 1 infrastructure is not allowed to consider as erroneous or invalid a mobile station simply because it has included a "SS version indicator" information element in one of the following messages : ALERTING, CONNECT, DISCONNECT, FACILITY, RELEASE, RELEASE COMPLETE and SETUP.

5.2.6 Information elements.

5.2.6.1 "Mobile station classmark".

A new value is used for the "revision level" field of the "mobile station classmark 1 and 2" information elements to indicate that the mobile station supports the 04.08 protocols defined for phase 2; this value was marked as "reserved for future use" in the phase 1 specifications.

Moreover, this "revision level" field has been reduced from 3 bits to 2 bits. The bit which has been freed is marked as spare for phase 2. The remaining bits are still used to indicate the revision level of the mobile station.

On reception of a "mobile station classmark" information element with the "revision level" field set to one of the two following values : 000 or 001, an upgraded phase 1 infrastructure is not allowed to consider this information element as invalid and shall process the message which contains this "mobile classmark" information element.

In addition, it would be advisable that whatever the value of the "revision level" field an upgraded phase 1 infrastructure should not consider this information element as invalid and should process the message containing the "mobile station classmark" information element.

This case is one of the exceptions to section 5.1.2 of this ETR.

The "frequency capability" field of the "mobile station classmark 2" information element has been reduced from 3 bits to 1 bit. The 2 bits which have been freed are marked as spare for phase 2. The remaining bit is used to indicate whether a mobile station supports the extension band for GSM.

An upgraded phase 1 infrastructure is not allowed to reject a "frequency capability" field which contains value 001 .

Moreover the upgraded phase 1 infrastructure should assume that the mobile station supports band number 0.

This case is one of the exceptions to section 5.1.2 of this ETR.

The "encryption algorithm" field of the "mobile station classmark 1 and 2" information elements has been reduced from 2 bits to 1 bit. The bit which has been freed is marked as spare for phase 2. The other bit is used to indicate whether the mobile station supports the standard A5 encryption algorithm.

An upgraded phase 1 infrastructure should only check in the "encryption algorithm" field the bit which indicates support of the standard A5 algorithm. It is up to the upgraded phase 1 infrastructure to decide how to treat mobile stations which do not indicate the support of the standard A5 algorithm.

This case is one of the exceptions to section 5.1.2 of this ETR.

5.2.6.2 "Repeat indicator".

A clarification about the use of the "repeat indicator" information element has been added in phase 2. The purpose of the "repeat indicator" information element is to indicate how repeated information elements shall be interpreted when included in a message. The "repeat indicator" information element is included before the first occurrence of the information element which will be repeated in a message.

An upgraded phase 1 infrastructure shall include the "repeat indicator" information element only in those cases identified in the phase 2 specifications, namely: when two "bearer capability", two "low layer compatibility", or two "high layer compatibility" information elements are included in a message.

5.2.6.3 "Starting time".

For phase 2 a clarification has been made about the treatment of the "starting time" information element. The time interval during which the mobile station considers that the starting time has not been yet reached now lasts 32024 frames (instead of 42431 frames for phase 1) from the reception of the message which contained the "starting time" information element.

An upgraded phase 1 infrastructure must take into account the fact that if it sends a starting time that adds a substential number of frames (32024 frames + the number of frames for transmission of the message) to the current frame number, a phase 2 mobile station will consider this starting time as having already been reached.

5.2.6.4 "Mobile Identity"

For phase 2 the option of having a TMSI of variable length has been deleted, in phase 2 the length is always 4 octets.

An upgraded phase 1 infrastructure is not allowed to send a mobile identity information element containing a TMSI with length of less than 4 octets.

5.2.7 Use of phase 1 spare bits.

Phase 2 changes which are presented under this section are those which have made use of phase 1 spare bits; for these changes the general rule specified in section 5.1.1 of this ETR has to be applied.

In the "location updating type" information element, a bit marked as spare in phase 1 is now used to indicate whether a follow-on request is pending at the mobile station, for example to speed up call setup after the mobile station has been switched on.

In the "mobile station classmark 2" information element, two bits marked as spare in phase 1 are now used as a "SS screening indicator" field to indicate some supplementary service attributes.

In the "mobile station classmark 2" information element, a bit marked as spare in phase 1 is now used to indicate that the mobile station supports the pseudo-synchronized handover procedure.

The last octet of the "mobile station classmark 2" information element has been modified to indicate which encryption algorithms are supported by the mobile station. This octet was marked as spare in the phase 1 specifications.

In the "bearer capability" information element, a bit marked as spare in phase 1 is now used to indicate the 6 kbit/s radio interface data rate capability and a request for a further negotiation of an intermediate data rate.

6. A / Abis interface.

6.1 A interface.

No cross phase compatibility problems are found for the A-interface.

6.2 Abis interface.

No cross phase compatibility problems are found for the Abis-interface.

7. Supplementary Service handling.

7.1 General Aspects.

Due to MAP version negotiation, a supplementary service interacting with mixed phase infrastructure will operate according to phase 1.

Unused bits within ASN.1 components parameters shall be set to zero by an upgraded phase 1 infrastructure.

Supplementary service subscription information shall be stored against Elementary Basic Service Groups (EBSG) in HLRs and VLRs in an upgraded phase 1 infrastructure.

Individual basic service codes shall not be used for subscriber data management in an upgraded phase 1 infrastructure.

The following BSG Codes shall not be used in an upgraded phase 1 infrastructure:

- MHS,
- Videotext,
- Teletext,
- 3.1 kHz speech.

The alternate and followed by service codes are not used in phase 1 infrastructure. However, if an upgraded phase 1 infrastructure supports, the procedures for application of supplementary services to alternate and followed by services, these procedures shall be as specified in the phase 2 specifications.

7.2 Service Aspects.

7.2.1 Call Forwarding Services.

In the result of call forwarding interrogations the registration and activation bits in SS-status shall be set either to 1,1 or 0,0, ie 'registered and activated' or 'deregistered and deactivated' respectively, by upgraded phase 1 infrastructure.

7.2.2 Call Barring Services.

No upgrades to a phase 1 infrastructure are required.

7.2.3 Unstructured SS Data.

No upgrades to a phase 1 infrastructure are required.

7.2.4 Password Handling.

The parameters badPW-TryAgain and badPWFormat-TryAgain shall not be sent over the air interface by an upgraded phase 1 infrastructure.

Phase 1 infrastructure runs a timer during the register password procedure lasting 15 seconds. In an upgraded phase 1 infrastructure the value of this timer should be increased. Refer to the phase 2 specification.

History

Document history	
October 1993	First Edition
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